

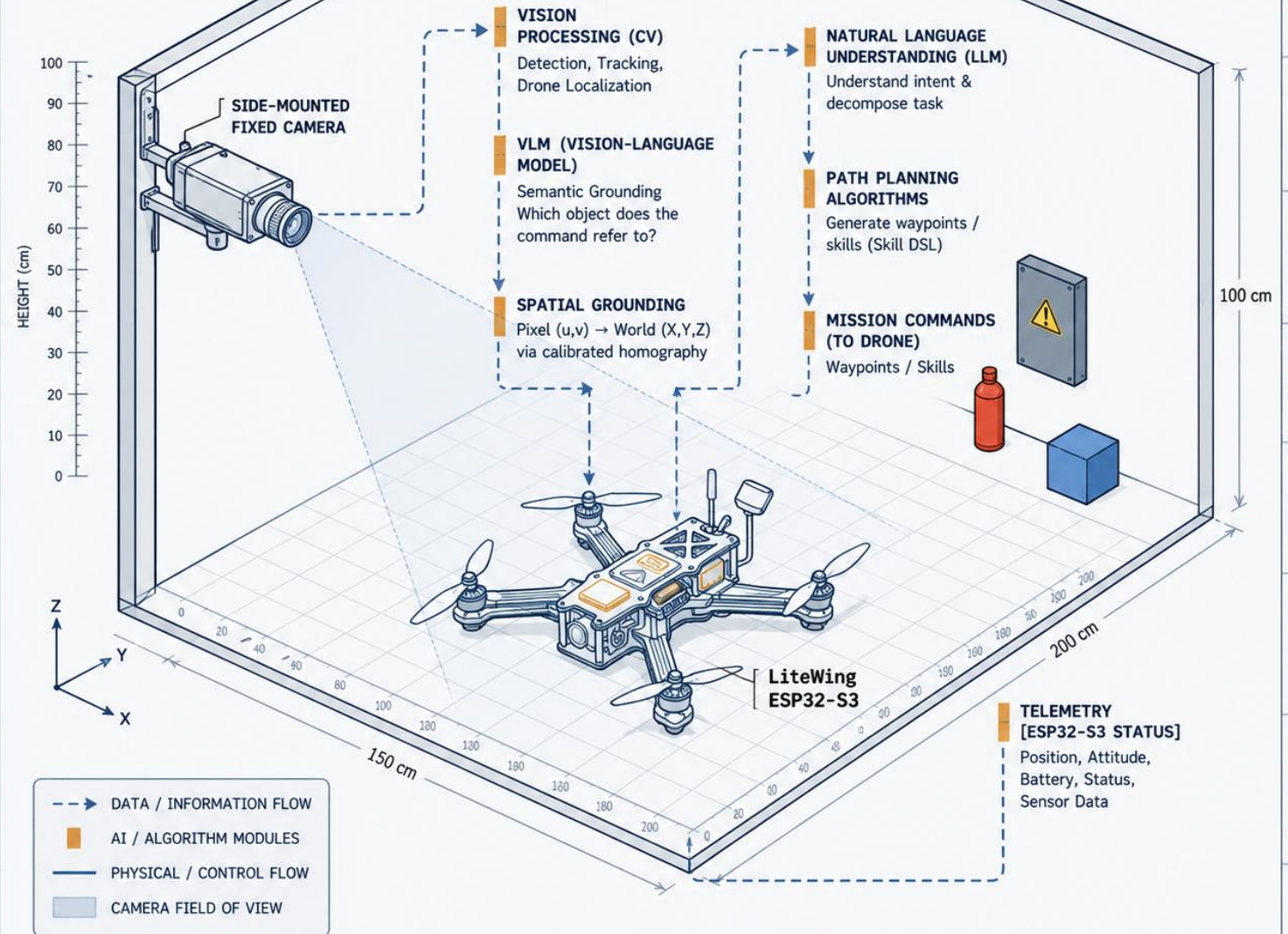
AERIS-Framework for Autonomous Drone Operation Using LLMs and VLMs

Natural-Language-Driven Indoor UAV Mission Execution

Domain: Robotics, Embodied AI
Project Stage: B.Tech Project Phase-1
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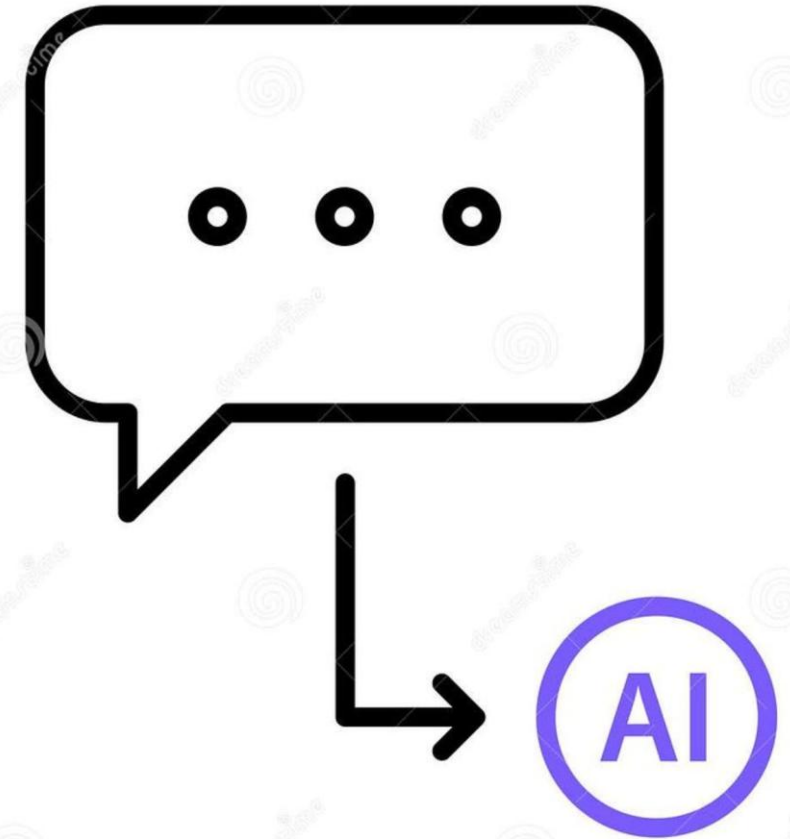
Introduction: Natural Language as the Drone Interface

Traditional operation uses joysticks, coordinates, or precomputed trajectories. We let users give high-level natural-language mission goals such as:

- “Fly in a square.”
- “Go to the blue object.”
- “Inspect the object beside the panel.”
- “Find the target and return.”

Pipeline (concept): USER LANGUAGE ↓ LLM (interpretation) ↓ Structured DRONE SKILLS ↓ INDOOR UAV EXECUTION

Note: The framework architecture is general, but our experimental application is bounded indoor UAV missions using an external fixed camera and a LiteWing ESP32-S3 platform.

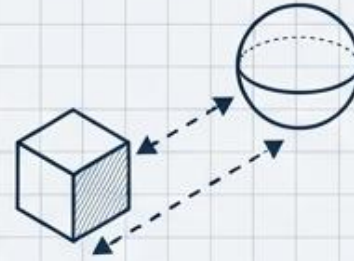


Motivation



1. Natural Interaction

Users should express mission goals in contextual natural language rather than calculating and inputting rigid physical coordinates or low-level commands.



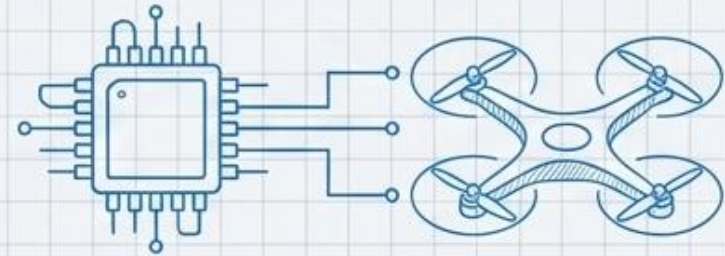
2. Semantic Understanding

Real environments consist of dynamic objects and relationships (e.g., 'beside the panel') that cannot be effectively represented by fixed spatial waypoints.



3. Adaptability

Precomputed missions and rigid trajectories frequently become invalid or dangerous the moment an environment dynamically changes.



4. Low-Cost Embodied AI

Exploring whether modern LLM/VLM reasoning can be decoupled from heavy hardware to direct a lightweight, low-cost ESP32-S3 drone instead of a high-end platform.

Objectives

1

1. NL → Structured Skills

Translate natural language into a compact, validated Skill DSL for drones.

2

2. Clean AI / Control separation

Keep high-level reasoning (LLM/VLM/planner) separate from deterministic flight control (PID/state estimator).

3

3. External perception & spatial grounding

Use a fixed camera with calibration/homography to convert pixels → world coordinates.

4

4. Safety & Skill DSL

Safety validator / geofence checks before execution; reusable skill primitives.

5

5. Feedback-driven replanning

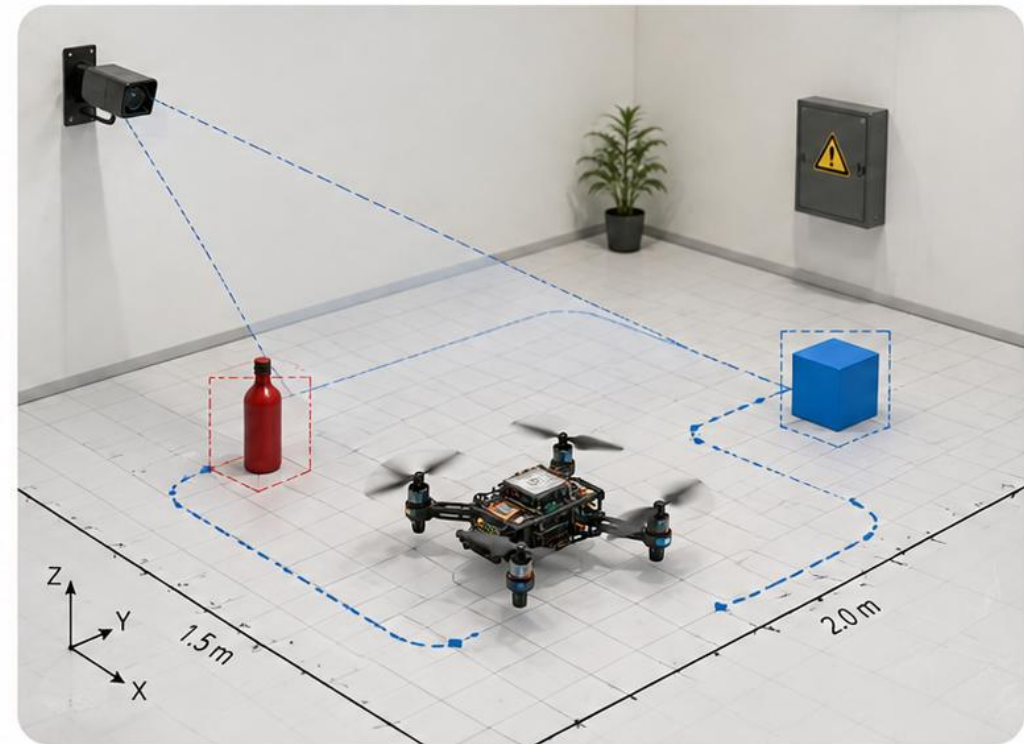
Enable mission replanning when mission-relevant state changes are detected.

Problem Statement

How can an indoor UAV execute high-level natural-language missions in a bounded workspace while grounding language-referenced targets visually and maintaining a safe separation between AI reasoning and real-time flight control?

Technical challenges

- **LANGUAGE** — Ambiguity and open-ended user expressions.
- **PERCEPTION** — Determine which visual entity matches the instruction.
- **SPATIAL GROUNDING** — Convert pixel observations into physical coordinates reliably.
- **EXECUTION & ADAPTATION** — Safety-validated actions and responsive replanning when environment changes.



Literature Survey & Research Gap

Paper	Summary / Relevance / Gap
[1] G. Chen et al., TypeFly, IEEE Trans. Mobile Comput., 2025	Focus: LLM-based low-latency UAV planning. Relevance: motivates structured high-level language-to-flight planning. Gap: does not address external-camera semantic grounding and lightweight indoor hardware constraints.
[2] X. Wang et al., Towards Realistic UAV VLN, arXiv:2410.07087, 2024	Focus: UAV vision-language navigation benchmarks and methodology. Relevance: emphasizes multimodal perception for UAVs. Gap: target platforms are higher-end; our work focuses on a modular stack for an ESP32-S3 indoor platform with external-camera grounding.

Research Gap: Existing work shows LLM planning and VLN research; our contribution is a lightweight, modular indoor UAV framework that explicitly connects Language → Semantic Visual Grounding (VLM) → Spatial Grounding → Safe Skills → Execution → Replanning.

Why VLM?

Conventional CV

Fast • Deterministic • Continuous tracking • Geometric outputs (WHERE in pixels). Example: red-region or YOLO bounding box detection for a known object.

VLM (Vision-Language Model)

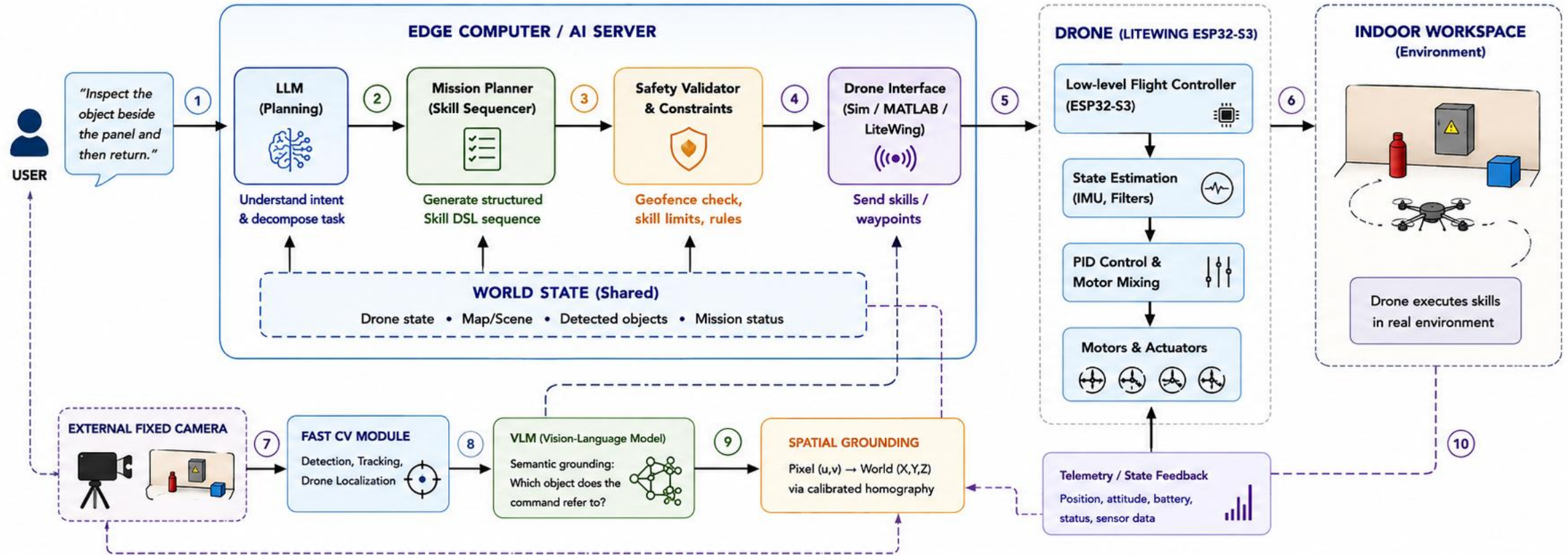
Multimodal (image + language) • Semantic reasoning • Open-vocabulary labels • Relational grounding • Answers WHICH object matches the instruction and visibility/semantic questions. Outputs identity + bounding box + pixel coordinates (not metres or motor commands).

Examples where VLM matters:

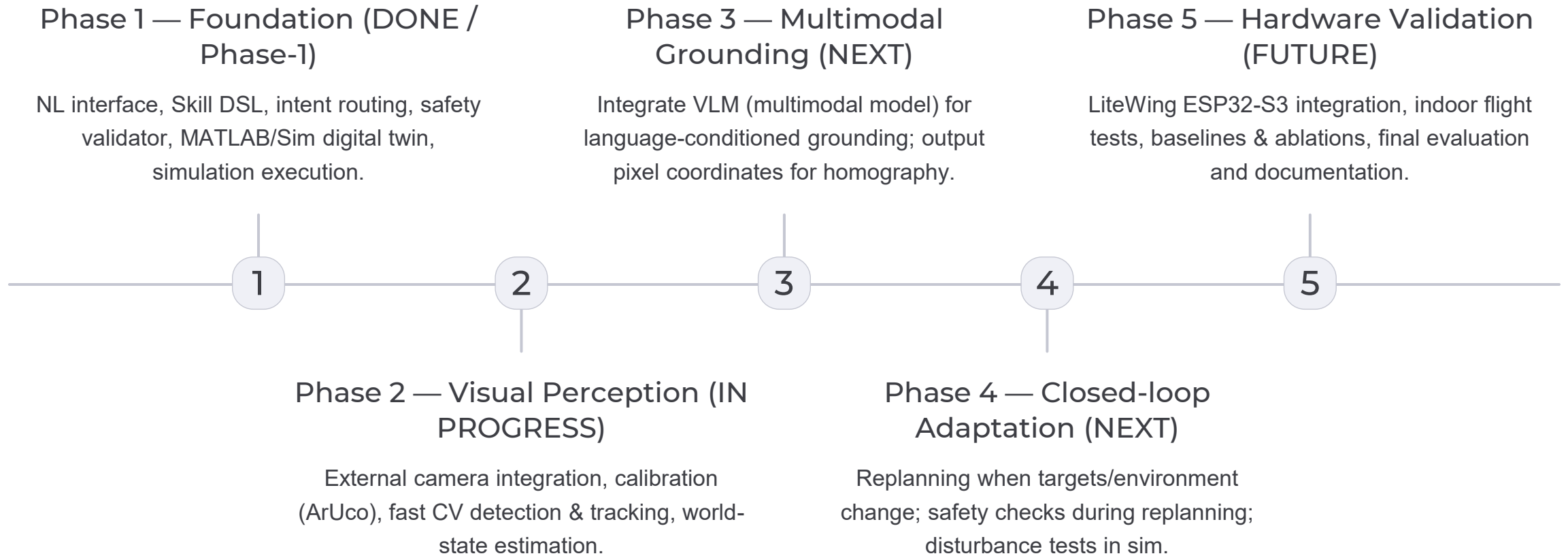
- “The bottle used for medicine” — semantic intent, may require label or text on object.
- “The red object beside the blue cube” — relational grounding (spatial relation between objects).
- “Inspect the object near the warning sign” — context-driven selection.

Conceptual separation (concise): LLM = WHAT to do • VLM = WHICH (visible semantic match) • CV = WHERE in pixels • Homography = WHERE in metres • Planner = HOW to sequence • Safety = IS IT ALLOWED? • PID/controller = HOW TO FLY THERE

Proposed Architecture & Methodology



Time Plan & Development Roadmap



📋 Status labels indicate relative progress. No claim that camera→VLM→closed-loop on physical drone is complete; it is the next integration milestone.

Current Phase-1 Implementation Status: Simulation-to-Real Transfer

Next Milestone (Target)

External
Camera

VLM Semantic
Grounding

Phase-1: Completed & Validated

Natural-Language
Input Parsing

LLM Translation
into Structured
Skills

Safety / Geofence
Validation
Interceptor

MATLAB / Simulink
Digital Twin
Execution Loop

Why Simulation First?

Simscape and Simulink model the LiteWing's dynamics. This allows us to safely validate high-level LLM reasoning and low-level controller parameters before risking the physical ESP32-S3 hardware.

References

- [1] G. Chen, X. Yu, N. Ling, and L. Zhong, “TypeFly: Low-Latency Drone Planning With Large Language Models,” *IEEE Transactions on Mobile Computing*, vol. 24, no. 9, pp. 9068–9079, 2025.
- [2] X. Wang, D. Yang, Z. Wang, H. Kwan, J. Chen, W. Wu, H. Li, Y. Liao, and S. Liu, “Towards Realistic UAV Vision-Language Navigation: Platform, Benchmark, and Methodology,” *arXiv preprint arXiv:2410.07087*, 2024.
- [3] Y. Emami et al., “Large Language Model-Assisted UAV Operations and Communications: A Multifaceted Survey and Tutorial,” *arXiv preprint arXiv:2602.19534*, 2026.
- [4] H. Chen et al., “Vision-and-Language Navigation for UAVs: Progress, Challenges, and a Research Roadmap,” *arXiv preprint arXiv:2604.13654*, 2026.
- [5] M. Krupáš, L. Urblík, and I. Zolotová, “Multimodal AI for UAV: Vision–Language Models in Human–Machine Collaboration,” *Electronics*, vol. 14, no. 17, p. 3548, 2025.
- [6] L. Yuan, C. Deng, D.-J. Han, I. Hwang, S. Brunswicker, and C. G. Brinton, “Next-Generation LLM for UAV: From Natural Language to Autonomous Flight,” *arXiv preprint arXiv:2510.21739*, 2025.

